

Game Master rulebook

Run the mission fairly, explain the results and retain the learning evidence.

The shared mission

All six guilds build one supply chain. In five rounds, ship at least 3 Titans (very easy), 7 (moderate) or 9 (very hard). Only Titans approved by Inspection count. Every guild must keep at least 1 gold, including after final settlement. Default: 10 gold per guild.

Set up one shared classroom game

Use ONE host browser and project it. Split the class into six guilds; choose a spokesperson for each. Other devices can read the guides, but do not share the live mission. The role menu is not a login.

Select France or Kazakhstan and the display language. Keep the Game Master role. Choose a preset, session name and starting gold, then "Start new mission". Start with 3 Titans for a first lesson. Use a separate practice run to demonstrate the screen.

Optional: use "Learning records -> Before play" after starting but BEFORE the first supplier vote. Ask students to use aliases. Default missions last five rounds with 10 gold per guild; custom settings change the challenge.

Follow the production order

Frames -> Engines -> Armour -> Subassembly -> Final assembly -> Inspection. The game follows this order once per round. Intermediate units stay in stock until the next stage uses them; they are not yet shipments.

Run this loop in every round

VOTE. Collect one vote each from Frames, Engines and Armour. Enter all three and a reason; lock the contract. Delivery is automatic. Never ask for a supplier D10.

TAKE SIX TURNS. Follow Frames -> Engines -> Armour -> Subassembly -> Final assembly -> Inspection. Each guild locks its prediction and reason. Then follow the displayed diagnostic or remedy, if any. You may enter agreed decisions as Game Master or hand over the role menu.

CHECK, THEN RELEASE. Pause for a recovery agreement or technician before release. Switch back to Game Master and click "Release good units" for EACH guild, even if its output is zero. This advances to the next guild.

REVIEW. After Inspection, compare shipments with the target and check gold plus unpaid bills. Ask: "What limited shipments? Who paid? What should change?" Arrange needed transfers BEFORE "Begin next round".

Supply: three votes before delivery

At the start of each round, Frames, Engines and Armour each vote; at least two votes choose the contract for all three. Standard: no assurance premium; 30% supplier-incident risk. Assured: 1 gold from EACH component guild (3 total); 10% risk. Each needs at least 2 gold to pay and survive. The computer resolves delivery automatically. Normally each receives 2 new raw units. A delay delivers 1 now and 1 next round; a quality incident places a defect in one incoming raw batch. Delayed units from earlier rounds arrive in addition to new receipts. Supplier risk never changes machine risk.

Routine machine risk is automatic

For an eligible turn without prevention: 10% major fault, 10% minor fault, 80% no fault. A turn already at zero for a known reason skips this check. Active verified prevention cancels a fault from the matching cause; it does not replace it with another fault.

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Follow the displayed action. Keep at least 1 gold after every payment.

Dice: let the game tell you

No production D6 and no supplier D10. If your initial feasible output is 1 or 2, skip D10. Known zero output (missing inputs, scheduled work or a known quality restriction) also skips D10. Only a new unexplained zero-output stoppage gets ONE free D10: 1-5 inconclusive; 6-10 identifies the existing cause. No reroll, extra cost or new fault. Physical D10: 0 means 10. A later repair may restore output without another D10.

One technician for the whole factory

Before release, buy the technician for 2 gold when the button is available: recover up to 4 eligible work hours. There is only ONE service slot per round across all six guilds. It cannot create materials, exceed 2 units, cancel a hold or remove Inspection's extra rework time. Compare output with and without the service before paying.

Support another guild with a clear promise

In "Recovery agreements", a sponsor offers 2 gold to the active guild for a technician, plus a quantity (1 or 2 units), deadline and reason. Acceptance transfers the gold and buys the service together. The sponsor must retain 1 gold. The promise measures the recipient's TOTAL new output after acceptance, not the extra output caused by support. It is best-efforts: the game records met/missed; there is no automatic penalty or repayment. Unaccepted offers expire when that guild releases.

Facilitator controls: use sparingly and explain

Game Master alone starts/resumes/imports missions, enters the common vote, releases units, advances rounds, settles and exports. During play, "Game Master intervention" can adjust gold, add/remove 4 or 8 work hours before a guild releases, or record a note. A note changes no calculation; every intervention marks the mission "assisted". Use Undo only to correct an entry, with a reason, never to obtain a better roll. A closed game cannot accept normal new operating actions.

Close the mission: settlement, not a sixth round

After round 5, click "Settle final obligations" to open settlement, check remaining bills and arrange any permitted support, then click it again to finish. There is no extra production. Unpaid deferred-repair invoices are paid once; already-paid work is not charged twice. Success requires the target AND every guild above zero. Outstanding hours and commitments are reported, not extra production opportunities. Reaching the target early does not end the remaining rounds.

Assess reasoning, not luck or wealth

The instructor rubric scores four dimensions 0-4 each: operational reasoning, decision quality, coordination, and reflection/transfer. Each has 25% weight. It is a teaching rubric, not a validated test. There is no richest-guild winner in 4.1. A good decision may still have an unlucky outcome; a transfer is not automatically good coordination.

End with evidence, then save

Use "Ledger & history" to compare predictions with actual output and costs. In "Learning records", add reflections and after-play tasks; a later transfer task is for a later class, not a scheduled reminder. After the game, "Debrief & alternatives" compares one recorded decision without changing the live result. Save game, export ledger CSV and export learning report as Game Master. Use aliases, not sensitive student details.

Use the 4.1 rules, not the old dice table

These rules apply to both country pages. Country changes names and artwork, not mechanics. This is a shared-browser teaching simulation, not secure multiplayer. Routine risks are computer-generated; the hidden event records are not secure from someone inspecting a save or the code. Use version 4.1 saves only. Old 15-round instructions, production D6 rules and vendor-change counters do not apply.

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Dice and production: use the rules for the version you are playing.

What is one unit?

One unit means one frame, engine, armour component, subassembly, assembled Titan or approved Titan, depending on the guild. A frame is not a shipped Titan. Only Inspection-approved Titans count towards the mission. In version 4.1, output is calculated from available hours and complete input sets; it is NOT assigned by a production D6.

Production: 0, 1 or 2 units

0 units = no complete output is released by this guild. A required input may be missing, fewer than 4 usable hours may remain, a repair may occupy the day, or a hold may block release. Example: 2 frames + 0 engines = 0 subassemblies, even with 8 hours. A known shortage does not trigger D10; only a new unexplained zero-output stoppage does.

1 unit = one complete output unit is released, not a half-finished unit. An ordinary production stage may have 4-7 usable hours and at least one input set, or time for two units but inputs for only one. Example: 2 frames + 1 engine and 8 hours give 1 subassembly; the unused frame remains in stock.

2 units = two complete outputs: full normal daily capacity. An ordinary stage needs 8 usable hours, two complete input sets and no hold. Example: 2 frames + 2 engines and 8 hours give 2 subassemblies. Extra materials alone cannot raise the limit above 2.

Inspection exception: a clean Titan needs 4 hours. A defective Titan needs 8 hours including rework, plus 1 gold, and Inspection must retain 1 gold. Thus 8 hours can approve two clean Titans OR one defective Titan after rework. QA Hold keeps defective Titans waiting but does not stop clean Titans from shipping. Displayed output is the final release after these checks.

D6: what every result means in version 4.1

D6 means a six-sided die numbered 1-6. Roll ONCE only when verification is requested after corrective action. There is no production D6 in version 4.1 and no face 0. A verification result is not a count of units produced.

Roll 1: no future prevention is established. The current repair still takes place; its charge and work hours remain. This does NOT automatically mean zero production today.

Roll 2, 3, 4, 5 or 6: verification succeeds. Prevent the matching fault cause for the next three rounds. Other causes remain possible; protection does not stack. Output still depends on hours, inputs and quality release.

Example: a minor corrective repair uses 4 of the 8 hours. With enough inputs and no other constraint, output can be 1 unit whether D6 is 1 or 5. The difference is future prevention, not today's unit count. Successful verification does not restore all 8 hours.

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Dice and production: use the rules for the version you are playing.

D10: what every result means in version 4.1

D10 is a ten-sided diagnostic die. The game requests ONE free roll for a new unexplained stoppage with initial feasible output 0. It is not a production or supplier-risk die. Do not roll for output 1 or 2, missing inputs, scheduled work or a known quality restriction.

Roll 1, 2, 3, 4 or 5: inconclusive diagnosis. The existing cause remains unknown; the roll adds no new fault, cost or downtime. No reroll. Choose a remedy; paid diagnosis is available, and corrective action includes a diagnosis charge if the cause is still unknown.

Roll 6, 7, 8, 9 or 10: the existing cause is identified. All five faces mean successful diagnosis, NOT five different faults. Identification alone does not repair the fault or add output. Corrective action now needs no extra diagnosis charge for that cause.

Physical D10 marked 0-9: 0 means 10, not zero production. The 4.1 input form accepts 0 and records it as 10. Example: D10 = 8 after an unexplained zero identifies the cause, but a remedy must still be chosen. D10 = 3 leaves it unknown without another roll.

D4: an older-edition die, not a version 4.1 action

D4 is a four-sided die numbered 1-4. Version 4.1 does NOT use it: choose Temporary repair, Diagnosis, Corrective action or Hold/defer repair on screen. In the older 15-round edition ONLY: 1 = Stop and investigate (4h / 1 gold); 2 = Root cause analysis (8h / 2 gold); 3 = Software / parts fix (0h / 0 gold); 4 = Rework (8h / 2 gold charged to the origin). In that older system, 1 and 3 apply only to failure or misalignment, 4 only to defective parts, and 2 to any non-vendor incident. Inapplicable results are rerolled there, not in 4.1. No D4 face represents a production quantity.